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| **Fiscal Year:** | FY 2013 | **Task Last Updated:** | FY 07/17/2013 |
| **PI Name:** | Hoffman, Larry Ph.D. | | |
| **Project Title:** | Synaptic plasticity in mammalian vestibular maculae | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Developmental biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Musculoskeletal Biology<br>(2) Neurobiology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
| **PI Email:** | lfh@ucla.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 310-825-9823 |
| **Organization Name:** | University of California Los Angeles | | |
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| **PI Address 2:** | Box 951624 | | |
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| **City:** | Los Angeles | **State:** | CA |
| **Zip Code:** | 90095-1624 | **Congressional District:** | 33 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2007 Space Flight: Bion-M1<br>NNH07ZTT001N |
| **Start Date:** | 08/01/2009 | **End Date:** | 07/31/2014 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 4 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Smith, Jeffrey | **Contact Phone:** | 650-604-0880 |
| **Contact Email:** | jeffrey.d.smith2@nasa.gov | | |
| **Flight Program:** | Bion/Shuttle | | |
| **Flight Assignment:** | STS-131 ; Bion M1<br>NOTE: End date is now 7/31/2014, per A. Chu/ARC and NSSC info (Ed., 4/30/2013) | | |
| **Key Personnel Changes/Previous PI:** | July 2013 report: Dr. Ivan A. Lopez is added as a co-investigator. | | |
| **COI Name (Institution):** | Zampighi, Guido ( UCLA School of Medicine )<br>Schweizer, Felix ( University of California, Los Angeles )<br>Lopez, Ivan Axel ( David Geffen School of Medicine at UCLA ) | | |
| **Grant/Contract No.:** | NNX09AL57G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | The nervous systems of animals, including man, have been found to exhibit a remarkable ability to adapt to changes in the ambient gravity environment. An understanding of these adaptations is critical to managing human exploration and survivability during long-term spaceflight. Previous studies have shown that a gravity-sensing structure within the inner ear, the utricle, exhibited quantitative changes in components of its constituent hair cells during microgravity exposures up to 14 days. This structure is known as one of the vestibular maculae, inner ear structures with important load sensing capabilities. The changes that were observed involved modification in the number of synaptic ribbons, which are organelles involved in the transfer of sensory signals from the hair cells to neurons that communicate with the central nervous system. Such changes represent a form of adaptation that is more generally characterized as synaptic plasticity. At present, however, it is not known whether these changes represented a transient effect of short-term spaceflight, or whether they are more permanent and reflect a chronic adaptive modification. Therefore, the present study is designed to test whether the approximately 30-day exposure to microgravity proposed for the BION-M1 satellite mission results in chronic modification in synapse number within hair cells of the utricle. This will be achieved through transmission electron microscopy and quantitative analyses to estimate the density of synaptic ribbons in different hair cell types and in different regions of utricles from flight and control animals. In addition, we will utilize new methods of conical tomography to visualize these ribbon synapses in three dimensions. This enables us to test whether synaptic plasticity is manifested in changes in subpopulations of ultrastructural elements of the synaptic ribbon. These results will be critical to a more comprehensive understanding of the long-term exposure to spaceflight in mammalian gravity-sensing systems. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research represents a high impact demonstration of the capabilities for vestibular hair cells to undergo synaptic plasticity due to alterations in the ambient gravitational field. This demonstration has led to a new investigation of hypergravity in which the mechanisms of this plasticity will be further explored. Eventually, these seminal investigations will segue to future studies of the molecular mechanisms underlying hair cell synaptic plasticity, which may potentially lead to the development of strategies for synaptic rehabilitation in pathologic conditions found in patients with compromised vestibular sensation. Such applications may be of significant value for vestibular rehabilitation following Meniere's disease or ototoxic damage due to chemotherapy. |
| **Task Progress:** | The principal tasks achieved during this funding year (August 1 2012 – July 31 2013) were: 1) we greatly improved our capabilities for transmission electron microscopy, conical tomography, and synapse reconstruction; 2) Drs. Hoffman and Schweizer developed and submitted a successful extension of this project in a ground-based study of utricular hair cell synaptic plasticity using hypergravity; and 3) we successful harvested and initiated analyses of specimens flown aboard the Russian BION-M1 biosatellite.<br>Due to the protracted temporal course of events involved in this flight project, it became necessary to recruit another investigator to complete the electron microscopy tasks involved in this project. Dr. Ivan Lopez (Adjunct Professor, Department of Head & Neck Surgery, UCLA) is now conducting all transmission electron microscopy (TEM) for this project. Dr. Lopez is a highly experienced microscopist and significantly elevates the capabilities of our investigator team. Based upon his extensive experience in conducting TEM analyses of inner ear tissues, he made critical improvements to the fixation and staining of specimens for ultrastructural analysis. He has developed critical expertise in obtaining conical tilt TEM series of utricular hair cell synapses, and is currently processing all specimens. He is also in optimal position to expand our capabilities and use the 300kV TEM facilities at UCLA. This may potentially improve the efficiency of this task by enabling thicker sections to be examined, which enhances the capabilites for capturing complete presynaptic structures in one section. This significantly enhances our scientific capabilities toward the objectives of this project.<br><br>In addition, improvements have been made to the software that creates 3D recontructions from conical tilt TEM tomography image series of synaptic ribbons. This software is being integrated into ImageJ/Fiji for analysis of neurotransmitter vesicle pools and other ultrastructural analyses.<br><br>A new research proposal was developed and submitted in response to the NASA Research Announcement (NRA) NNH12ZTT001N (September 2012). The foundation for this new work, which represents a ground-based study of utricular hair cell synaptic plasticity, was the evidence for plasticity that was obtained in specimens harvested from the STS-131 Biospecimen Sharing Program, and therefore is firmly based upon a principal outcome of the current project. We will utilize hypergravity as the tool with which to alter the ambient gravitational field.<br><br>At long last, the BION-M1 biosatellite was launched on 19 April 2013, and landed on 19 May 2013. The entire team of Biospecimen Sharing Program (BSP) investigators was at the Institute of Biomedical Problems in Moscow for the postflight specimen harvesting. The BSP dissection flow was virtually identical to that which took place for the previous shuttle flights (STS-131, -133, and -135), with critical accommodations to meet the needs of our Russian scientific colleagues. Overall, the mission was an overwhelming success in terms of deliverables to the US BSP team. Analyses of temporal bone specimens are currently underway. |
| **Bibliography Type:** | Description: (Last Updated: 10/05/2022) |
| **Abstracts for Journals and Proceedings** | Hoffman LF. "Plasticity in Mammalian Gravitoinertial Receptors: An Adaptive Continuum?" Invited symposium presentation (Symposium III: Plasticity in the Gravistatic Vertebrate Inner Ear: Insights from Development and Evolution) at the joint meeting of the American Society for Gravitational and Space Biology and the International Society of Gravitational Physiology, San Jose, CA, November 2 - 6 2011.<br>Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 42. , Nov-2011 |